

Chun-Wei Chiang

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SUMMARY

Ph.D. Candidate in Computer Science focusing on Human-AI Interaction, Applied Machine Learning, and Large Language Models (LLMs), with hands-on experience in developing and deploying large-scale ML systems. Proven track record of building LLM-powered solutions that enhance AI-assisted decision-making and collaboration. Skilled in the end-to-end machine learning pipeline using Python, TensorFlow, and SQL. Published in top-tier venues such as CHI and WWW. Graduating in May 2025, I am seeking to leverage my ML expertise in building and improving innovative AI systems.

EDUCATION

Ph.D. in Computer Science , Purdue University	2020 - May 2025
M.S. in Computer Science , West Virginia University	2016 - 2018

PROFESSIONAL EXPERIENCE

Purdue University Graduate Research Assistant	January 2020 - Present
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- Designed and implemented an LLM-based chatbot to enhance human-AI teaming, improving collaboration performance by 13% in decision-making and also reducing decision bias.
- Led and conducted comprehensive qualitative and quantitative research, using A/B testing to assess human-AI team dynamics, resulting in enhanced usability and increased engagement.
- Built and deployed a full-stack web application using Django, Vue.js, and PostgreSQL, hosted on AWS Elastic Beanstalk, supporting data collection and analysis for over 4 research projects.
- Authored multiple peer-reviewed papers, demonstrating strong technical writing and coding abilities by developing and documenting research systems that led to published results.

Twitch Applied Science Intern	June 2023 - August 2023
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- Engineered and fine-tuned a sequential-based collaborative filtering algorithm, increasing item recommendation precision by 15% in pilot studies.
- Analyzed large-scale user behavior datasets using Python and SQL to optimize machine learning models and boost recommendation accuracy.
- Collaborated with cross-functional teams to integrate new algorithms into the production pipeline, enhancing system efficiency and scalability.

Honda Research Institute Research Scientist Intern	January 2023 - May 2023
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- Researched driver behavior and user experience in autonomous vehicles and conventional vehicles, leveraging quantitative analysis to design safer, more user-friendly interfaces.
- Analyzed large-scale driver interaction data with autonomous systems, identifying patterns that led to improved usability and increased driver satisfaction.

Brain Technologies Natural Language Processing (NLP) Research Intern	June 2022 - August 2022
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- Applied advanced NLP techniques and collaborative filtering algorithms to fine-tune and evaluate models for an AI-driven restaurant recommendation system.
- Improved existing recommendations using Large Language Models (LLMs), leveraging YouTube watch history data to enhance accuracy and relevance.
- Implemented a Retrieval-Augmented Generation (RAG) model by integrating GPT with online search technology to establish a dynamic knowledge base, enabling real-time processing of web information for context-aware responses with improved accuracy.

Covoir

January 2019 - August 2019

Co-founder

- Secured \$500K in seed funding, successfully leading the company's initial investment round, which enabled the transition from concept to operational startup.
- Spearheaded the development of a decentralized Oracle service for blockchain applications, providing real-time access to off-chain data and enhancing blockchain functionality.

West Virginia University

January 2017 - August 2019

Graduate Research Assistant

- Created a Google Chrome extension to collect 10K real-world data points on crowdsourcing tasks, analyzing worker's completion time and enabling better time estimation through data collection.
- Modeled task completion time, achieving approximately 70% prediction accuracy for hourly wages based on task content and metadata.
- Developed an online peer support tool to help crowd workers improve professional skills, resulting in a 32% increase in work efficiency and improving task completion rates across multiple online platforms.

Mitake Information

November 2014 - June 2016

Software Engineer

- Developed and maintained eight high-performance Android applications for stockbrokers, utilizing the Android SDK and ensuring smooth performance across devices. The apps reached over 1 million downloads, significantly enhancing user engagement and satisfaction in the financial services industry.

SKILLS

Machine Learning	Human-AI Interaction, Deep Learning, LLMs
Programming Language	Python, R, Java
Tools and Frameworks	Git, TensorFlow, Pytorch, Sklearn, Hugging Face
Database Processing	SQL, PostgreSQL, MySQL, MongoDB, Pandas, Data Visualization
Full-stack Web development	JavaScript, HTML, Vue.js, React, Django, Amazon Web Services (AWS)
Research Skills	Data Analysis, A/B Testing, Quantitative Research, Qualitative Research

HONOR AND CERTIFICATION

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- Google Certificate: Foundations of Project Management
 - Best Poster Honorable Mention, The World Wide Web Conference (WWW 2019)
 - Special Recognitions for Outstanding Reviews (DIS 2020, CHI 2024, CHI 2025)
 - Gary Marsden Travel Awards, Conference on Human Factors in Computing Systems (CHI 2023)
 - NSF Student Travel Award (IUI 2024, HCOMP 2018)

PUBLICATIONS

Published 10+ peer-reviewed papers on Human-AI interaction in top conferences, including CHI, IUI, and CSCW. [[Google Scholar](#)]

- [1] Chun-Wei Chiang, Zhuoran Lu, Zhuoyan Li, and Ming Yin. "Enhancing AI-Assisted Group Decision Making through LLM-Powered Devil's Advocate". In: *Proceedings of the 29th International Conference on Intelligent User Interfaces*. 2024, pp. 103–119.
- [2] Chun-Wei Chiang, Zhuoran Lu, Zhuoyan Li, and Ming Yin. "Are Two Heads Better Than One in AI-Assisted Decision Making? Comparing the Behavior and Performance of Groups and Individuals in Human-AI Collaborative Recidivism Risk Assessment". In: *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 2023, pp. 1–18.
- [3] Chun-Wei Chiang and Ming Yin. "Exploring the effects of machine learning literacy interventions on laypeople's reliance on machine learning models". In: *27th International Conference on Intelligent User Interfaces*. 2022, pp. 148–161.
- [4] Chun-Wei Chiang and Ming Yin. "You'd better stop! Understanding human reliance on machine learning models under covariate shift". In: *Proceedings of the 13th ACM Web Science Conference 2021*. 2021, pp. 120–129.